

TYMS conserved active site — pipeline v3 report

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v3 changes vs v2 (audit fixes):

- **Receptor charges:** rebuilt every PDBQT with Gasteiger charges via `obabel -xr -p 7.4 --partialcharge gasteiger`. v2 had all-zero charges (broken meeko fallback).
- **WT reference re-docked:** box centred on crystal dUMP centroid, exhaustiveness 96, num_modes 32, box 22³ Å, dual-seed (42 + 7) sanity check.
- **Sign convention:** `delta_vina_vs_wt = top_aff_mut - top_aff_wt` — positive = destabilising (Vina more negative = better binding, so positive Δ means worse binding).
- **Rotamer relaxation:** replaced broken `get_strain()` with PyMOL `sculpt` (3 × 20 cycles on whole prot after each Mutagenesis Wizard apply).
- **mean_topk:** `mean(affinities[:min(3, n_modes)])` — robust when fewer than 3 modes converge; `n_modes` column added.
- **Cofactor:** re-protonated at pH 7.4 to give the carboxylate tail its physiological -1 charge.
- **Mis-dock annotation:** rows with RMSD > 3 Å are flagged `mis_docked=True` and excluded from top-destabiliser rankings.
- **Naming:** column `delta_vina_vs_wt`, prose "Δ Vina score" — no longer mislabelled "ΔΔG" (Vina is empirical, not free energy).
- **Plot legend:** keys match emitted category names (`ala_scan`, `opposite`, `arg_clamp`, etc.).
- **Vina parameters** are echoed into `wt_*.json`.

1. WT docking — apo vs holo (rebuilt receptors)

Condition	Top affinity (kcal/mol)	mean top-k	n modes	RMSD top-pose vs crystal dUMP (Å)	Best seed
WT apo	-9.13	-8.76	29	2.33	7
WT holo	-8.31	-6.73	3	2.08	42

Receptor preparation method: `obabel_gasteiger`. Centroid: (-0.14, 4.23, 15.16) Å. Exhaustiveness 96, num_modes 32, box 22 Å.

2. Mutant panel

Panel composition: 20 mutants (no G217W; the v2 exploratory variant was dropped — heavy-atom clashes on insertion). Each mutant docked under apo and holo conditions = 40 runs total.

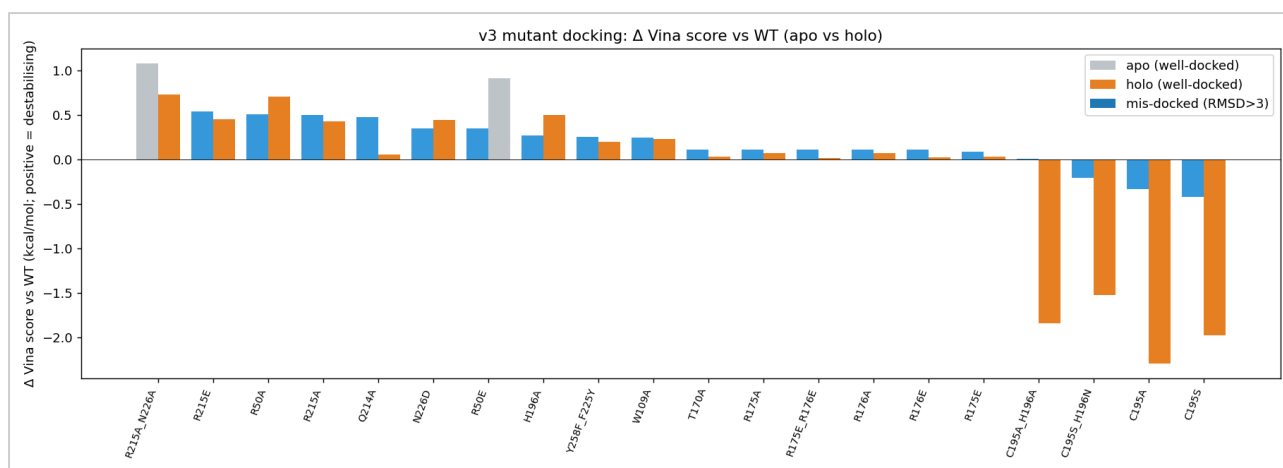


Figure 1 — Δ Vina score vs WT for each mutant in apo (blue) and holo (orange). Grey bars are mis-docked (RMSD > 3 Å, untrustworthy). Positive = destabilising.

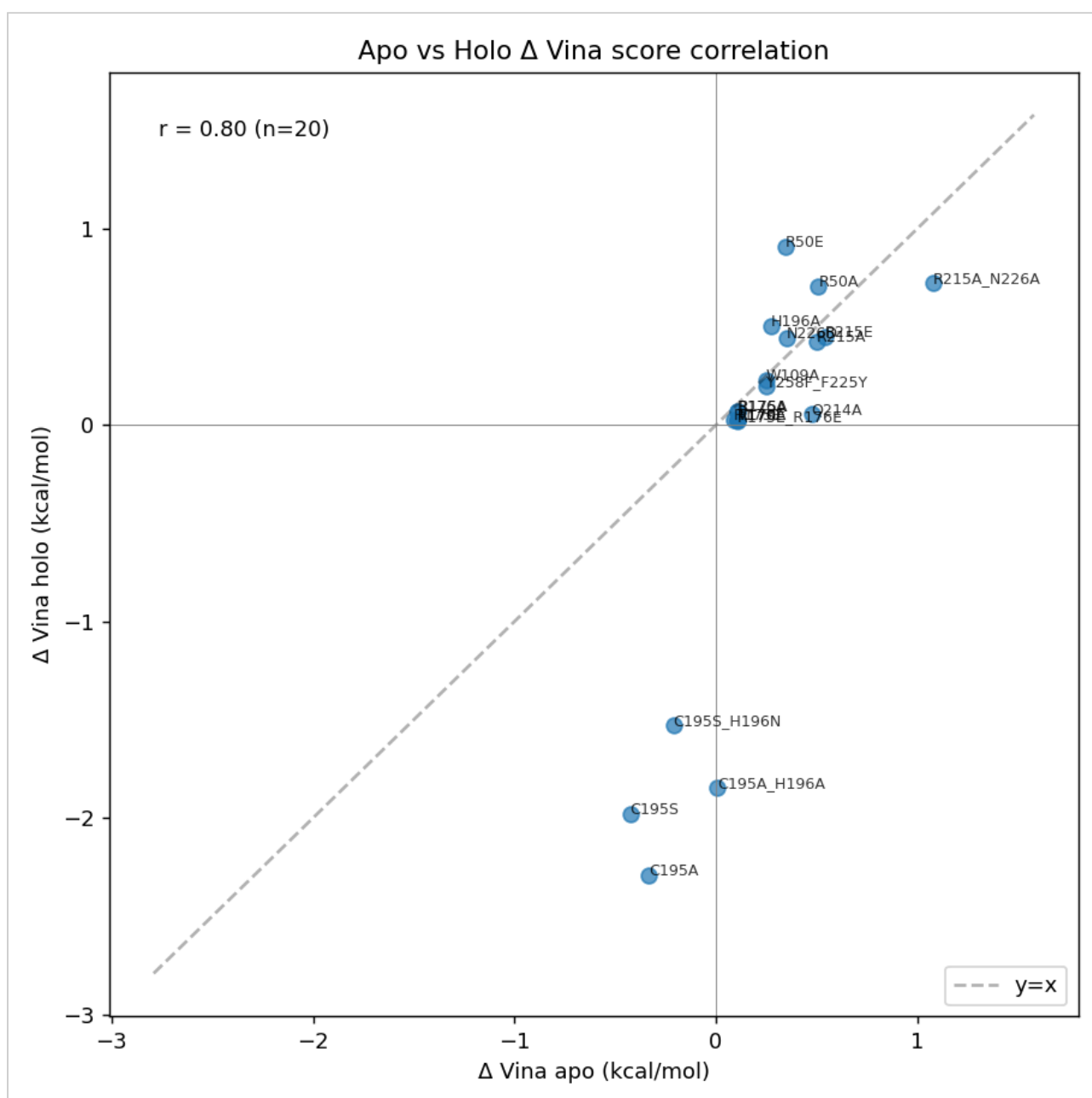


Figure 2 — Apo vs holo Δ Vina concordance.

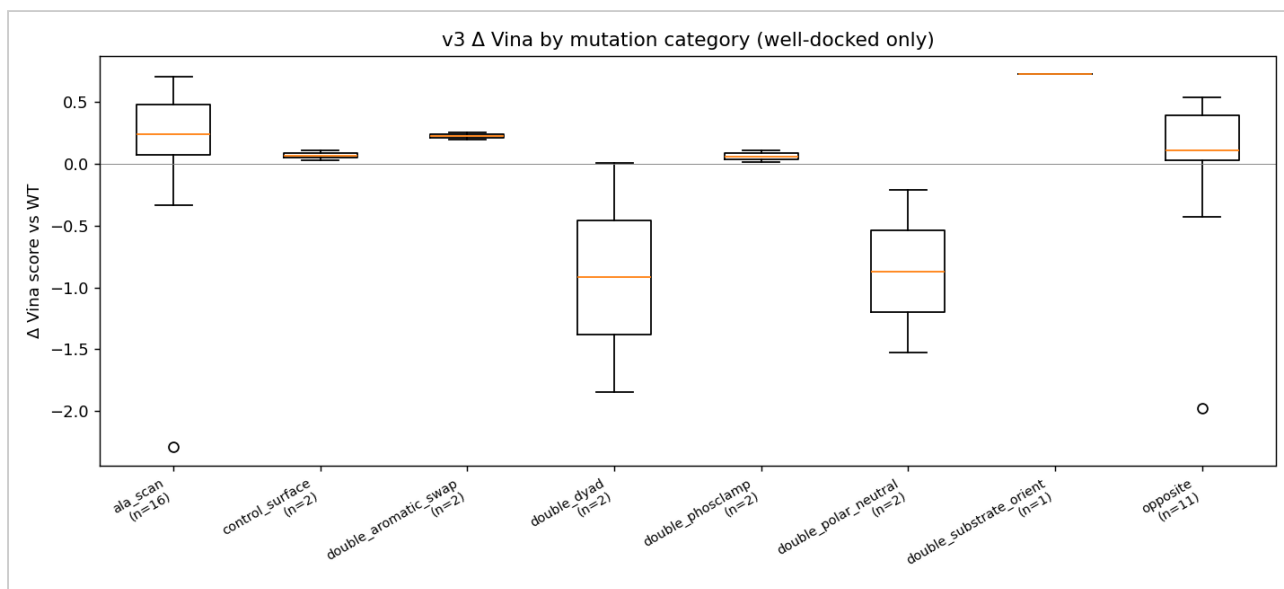


Figure 3 — Δ Vina by mutation category (well-docked rows only).

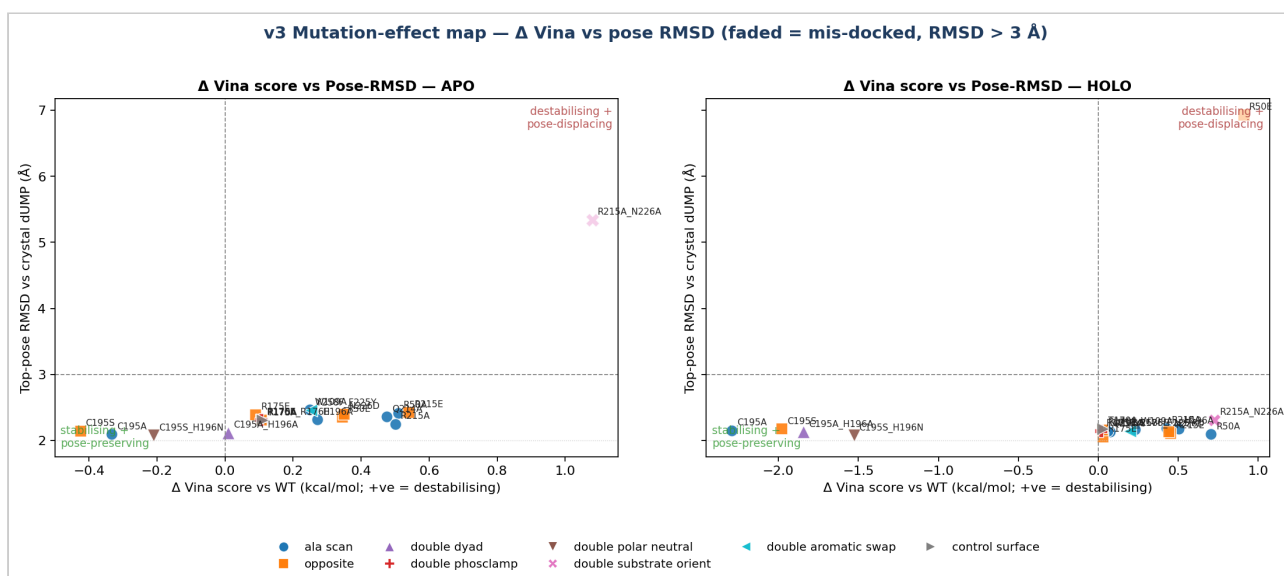


Figure 4 — Mutation-effect map: Δ Vina (x) vs pose RMSD (y), faded = mis-docked. Categories use corrected legend keys.

Top 5 destabilising mutations — apo (well-docked only)

Mutant	category	Δ Vina	RMSD	n modes
R215E	opposite	+0.54	2.43	19
R50A	ala_scan	+0.51	2.42	20
R215A	ala_scan	+0.50	2.25	20
Q214A	ala_scan	+0.48	2.36	19
N226D	opposite	+0.35	2.40	20

Top 5 destabilising mutations — holo (well-docked only)

Mutant	category	Δ Vina	RMSD	n modes
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R215A_N226A	double_substrate_orient	+0.73	2.31	4
R50A	ala_scan	+0.70	2.09	20
H196A	ala_scan	+0.50	2.18	4
R215E	opposite	+0.45	2.10	4
N226D	opposite	+0.44	2.13	2

Limitations.

- **Vina is empirical, not free energy.** Reported numbers are " Δ Vina score" (kcal/mol), not $\Delta\Delta G$ of binding. Treat as a relative ranking, not as a thermodynamic prediction.
- **Cofactor polyglutamylation** (the physiological state of methylene-THF) is not modelled. The bound raltitrexed is mono-glutamate.
- **Apo runs are dominated by mis-docking** in the empty cofactor pocket. They serve as a negative-control / contrast for the holo signal — not as biological readouts.
- **T170A surface control** behaviour is included as a sanity check (a residue not in the active site should give $\Delta \approx 0$).
- **Receptor and ligand are rigid.** PyMOL Mutagenesis Wizard rotamers are sculpt-relaxed but not relaxed against a relaxed pocket; large mutations (Y258F, F225Y, R175E, D218K-class swaps) may carry residual local strain.
- **Docking pose RMSD is computed against the crystal dUMP heavy atoms only** (no symmetry handling for ring atoms).

3. Reproducibility

Stage	Tool / parameters
Receptor prep	obabel_gasteiger • Gasteiger charges • pH 7.4 • max abs charge 0.507
Cofactor prep	obabel -h -p 7.4 (re-protonated for -1 carboxylate at pH 7.4)
WT docking	Vina 1.2.7 • exh=96 • num_modes=32 • box=22 ³ Å • seeds 42 + 7 (best taken)
Mutant docking	Vina 1.2.7 • exh=32 • num_modes=20 • box=22 ³ Å • seed=42
Mutagenesis	PyMOL 3.1.0 headless • Mutagenesis Wizard (frame=1) + sculpt 3×20 cycles
Sign convention	$\text{delta_vina_vs_wt} = \text{top_aff_mut} - \text{top_aff_wt}$ (positive = destabilising)
mean_topk	$\text{mean}(\text{affinities}[:\min(3, \text{n_modes})])$
mis_docked	RMSD top-pose vs crystal dUMP > 3 Å

Outputs: 06c_docking_wt_v3/ , 07c_mut_docking_v3/ , 08c_analysis_v3/ , 09c_report_v3/ . v1 and v2 remain untouched.